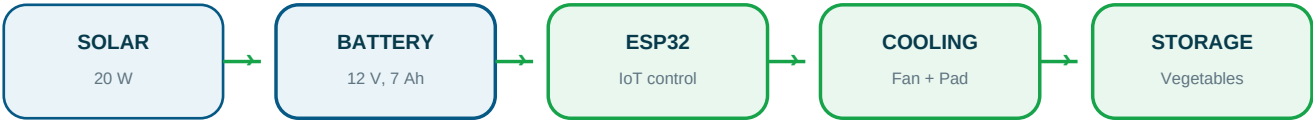


IoT Based Solar Powered Smart Mini Cold Storage

For Fresh Vegetables



Solar Energy • Natural Evaporative Cooling • IoT Monitoring

Student / Team	[ADD NAME]
Department	Electronics and Communication Engineering
College / University	[ADD COLLEGE NAME]
Academic Year	2026

A compact, solar-supported storage concept combining evaporative cooling with ESP32-based temperature and humidity monitoring.

Abstract

Fresh vegetables are highly perishable and can suffer quality loss when suitable storage is unavailable. The proposed project presents a compact smart mini cold-storage concept that combines solar energy, battery backup, water circulation, a honeycomb cooling pad, forced airflow and IoT-based environmental monitoring.

A 20 W solar panel is proposed as the renewable power source, with a 12 V battery for energy storage. A 12 V water pump circulates water to a honeycomb cooling pad, while a DC fan moves air through the wet pad to support evaporative cooling. An ESP32 controller receives temperature and humidity data from a DHT22 sensor and can display the values on a 0.96-inch I2C OLED. The design is intended as a low-power, compact and portable prototype for fresh-vegetable storage.

Scope: This report documents the proposed prototype design, architecture, components and operating principle. Quantitative cooling performance and shelf-life claims require controlled experimental validation.

Introduction

The project is motivated by the need for small-scale vegetable preservation in locations where conventional cold-storage facilities, dependable grid electricity or rapid transport may not always be available. A decentralized mini-storage unit can provide a local point for temporary storage while using renewable energy and low-power components.

The proposed system focuses on three connected ideas: solar-supported energy, natural/evaporative cooling and IoT monitoring. Together, these features create a platform that can be tested and upgraded for different crops, climates and storage capacities.

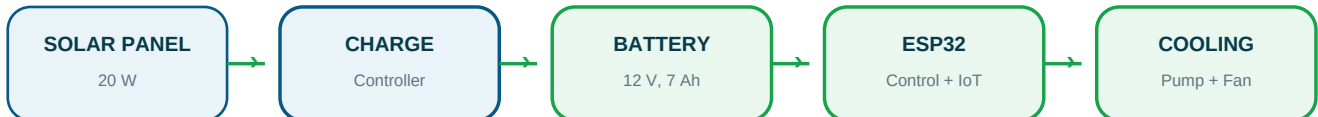
Problem Statement & Objectives

Problem statement: Design a compact, low-power and solar-supported vegetable storage system that can provide cooling and environmental monitoring without depending completely on conventional electricity.

- Develop a compact mini cold-storage prototype.
- Use solar energy as the primary proposed power source.
- Provide battery backup for system operation.
- Use water circulation and a honeycomb cooling pad for evaporative cooling.
- Monitor temperature and relative humidity using ESP32 and DHT22.
- Display sensor readings using an OLED.
- Explore a low-cost, portable design for small-scale use.
- Provide a base for future cloud and mobile monitoring.

Proposed System & Architecture

The system is organized into energy, cooling, sensing/control and storage sections. Solar energy is routed through a charge controller to the battery. The stored energy supplies the controller and 12 V loads. The ESP32 processes sensor data and can control the fan and pump through an appropriate switching stage.



The cooling section uses a 5 litre water tank, 12 V pump, water distribution pipe and honeycomb pad. Airflow from the DC fan passes through the wet pad and then toward the vegetable chamber. The DHT22 provides temperature and humidity measurements to the ESP32.

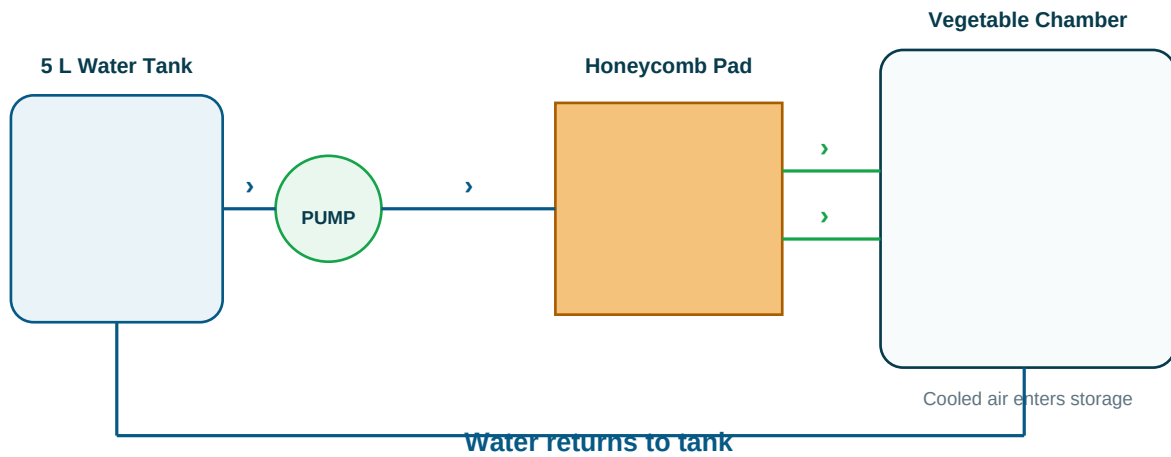
Physical Prototype Design

The reference prototype is a compact insulated box with an external size of 60 × 40 × 40 cm. The design includes an insulated lid, thermal insulation, food-grade inner chamber, front airflow arrangement, display, water tank and pump. A solar panel is shown on the top surface in the concept design.

Parameter	Specification
Outer length	60 cm
Outer width	40 cm
Outer height	40 cm
Thermal insulation	25 mm EPS / thermocol
Inner food-grade box	Approx. 55 × 35 × 35 cm
Vegetable chamber	35 × 28 × 25 cm
Cooling pad	30 × 20 × 5 cm
Water tank	5 litre

Internal / Cross-Section Design

The internal arrangement places the airflow and evaporative-cooling components above or beside the vegetable chamber while keeping the water reservoir and pump accessible. The exact physical arrangement should be matched to the final fabricated prototype.



Water path: Tank → Pump → Distribution pipe → Honeycomb pad → Return to tank.

Air path: Fan → Wet honeycomb pad → Cooled air → Vegetable storage chamber.

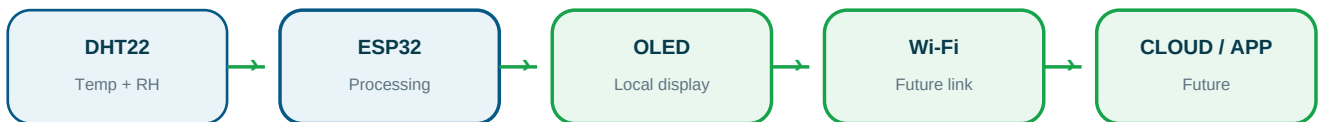
Cooling & Water Circulation

The proposed cooling method is evaporative cooling. Water is circulated over the honeycomb pad. When air moves across the wet surface, evaporation can remove heat from the passing air. The resulting airflow is directed toward the vegetable storage chamber.

- Water is stored in the 5 litre tank.
- The 12 V pump moves water toward the distribution pipe.
- The distribution pipe wets the honeycomb cooling pad.
- The DC fan drives air through the wet pad.
- The cooled airflow is directed into the storage chamber.
- Water is collected and returned to the tank for recirculation.

IoT Monitoring System

The ESP32 is the central controller for the proposed monitoring system. A DHT22 sensor measures temperature and relative humidity. The readings can be shown locally on a 0.96-inch I2C OLED. Because the ESP32 includes wireless connectivity, the design can later be extended to a cloud dashboard or mobile application.



IoT extension: The website associated with the poster can act as a project information dashboard. Live sensor data should only be presented as live data after the ESP32 is actually connected to a backend or suitable IoT service.

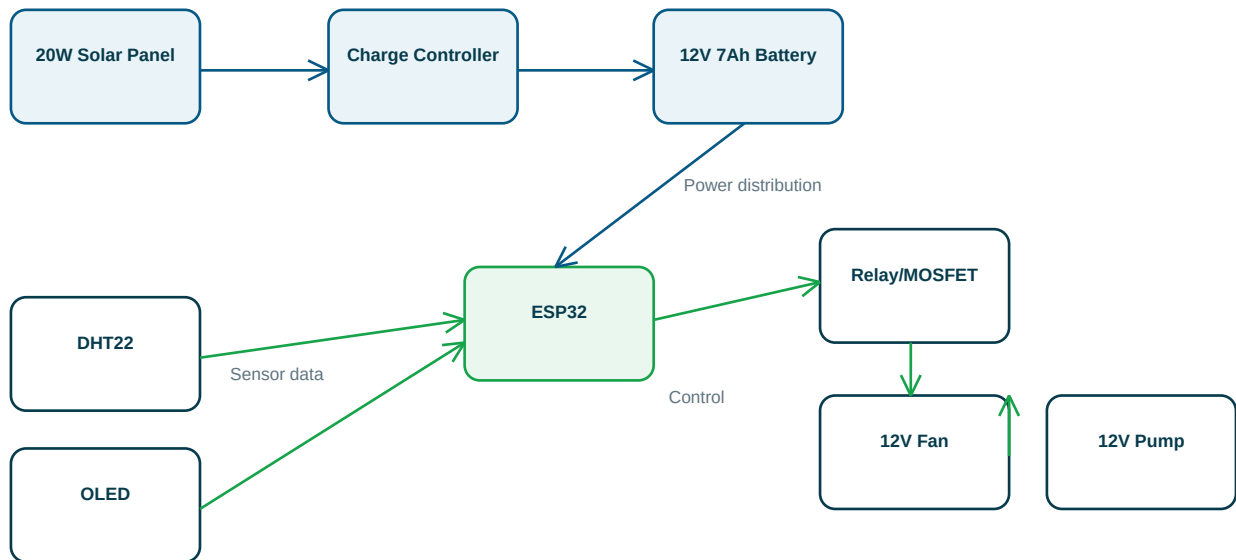
Components & Specifications

No.	Component	Specification / Size	Qty.
1	Outer plastic box	60 × 40 × 40 cm	1
2	Thermocol / EPS	25 mm thickness	1 set
3	Food-grade inner box	Approx. 55 × 35 × 35 cm	1
4	Honeycomb cooling pad	30 × 20 × 5 cm	1
5	DC fan	12 V, 120 mm	1
6	Water pump	12 V DC	1
7	Water tank	5 litre plastic	1
8	Solar panel	20 W	1
9	Battery	12 V, 7 Ah	1
10	ESP32 Dev Module	ESP32-WROOM-32	1
11	DHT22	Temperature + humidity	1
12	OLED	0.96-inch I2C	1
13	Relay / MOSFET	12 V load control	1 set
14	Wires / switch / connectors	As required	1 set

Circuit Diagram & Electrical Setup

The following is a conceptual electrical architecture for the project. Final wiring must be verified against the actual modules, their voltage/current requirements and the selected switching components. Exact ESP32 GPIO numbers are intentionally not specified here because they depend on the final tested wiring.

Conceptual circuit: exact GPIO pin numbers should be assigned from the final tested wiring.



Power: Solar panel → charge controller → 12 V battery. The battery supplies the 12 V fan and pump and the appropriate regulated supply for the ESP32/sensor/display.

Control: DHT22 → ESP32; ESP32 → OLED; ESP32 → relay/MOSFET → fan/pump.

Working Principle

- Solar energy is converted to electrical energy and used to charge the battery through a suitable charge controller.
- The battery provides stored energy for the low-power control system and 12 V loads.
- The pump circulates water from the tank to the cooling pad.
- The fan moves air through the wet honeycomb pad.
- The cooled airflow is directed toward the vegetable chamber.
- The DHT22 measures temperature and relative humidity.
- The ESP32 processes the measurements and presents them on the OLED.
- Future firmware can transmit data over Wi-Fi for remote monitoring.

Expected Performance & Observations

This project report distinguishes design targets from measured experimental results. The values below describe the intended operating range or design reference and should not be treated as validated field performance unless measurements are recorded under controlled conditions.

Item	Design / Expected Reference	Status
Temperature monitoring	Continuous DHT22 measurement	To be tested
Humidity monitoring	Continuous DHT22 measurement	To be tested
Solar input	20 W panel	Design value
Battery	12 V, 7 Ah	Design value
Water capacity	5 litre	Design value
Cooling method	Evaporative / honeycomb pad	Prototype concept
Long-duration storage	Requires crop-specific testing	Not validated

Actual test results: Experimental validation and long-duration performance testing are required before final quantitative claims about cooling efficiency, shelf-life extension or reduction in post-harvest losses can be made.

Advantages & Limitations

Advantages

- Solar-supported operation
- Low-power concept
- IoT monitoring capability
- Compact and portable
- Natural evaporative cooling
- Modular and upgradeable

Limitations

- Cooling depends on ambient conditions
- Evaporative cooling needs water
- Solar availability affects operation
- Prototype capacity is limited
- Field testing is required

Future Scope

Upgrade	Purpose
Mobile application	Remote monitoring and notifications.
Cloud IoT dashboard	Online storage of temperature and humidity data.
Remote alerts	Notify users when conditions cross selected limits.
Automatic control	Automatically control fan/pump based on measured conditions.
MPPT solar charging	Improve solar energy harvesting with a suitable controller.
Larger capacity	Adapt the design for higher vegetable storage volume.
AI-based freshness prediction	Explore crop-specific prediction using historical sensor data.
Optional additional cooling	Evaluate an additional cooling stage where evaporative cooling is insufficient.

Conclusion

The IoT Based Solar Powered Smart Mini Cold Storage is proposed as a compact platform that combines renewable energy, evaporative cooling and environmental monitoring. The design uses a 20 W solar panel, 12 V battery, 12 V pump, DC fan, honeycomb cooling pad, ESP32, DHT22 and OLED to create a modular prototype.

The project provides a practical base for experimentation and future development. Controlled testing is the next important step to determine actual temperature reduction, humidity behavior, energy consumption, water consumption and crop-specific storage performance.

The project can subsequently be expanded with cloud connectivity, mobile monitoring, automatic control and larger storage capacity.

References & Acknowledgement

1. Espressif Systems — ESP32 technical documentation.
2. DHT22 / AM2302 temperature and humidity sensor datasheet.
3. 0.96-inch I2C OLED display module documentation.
4. Solar photovoltaic system design and installation references.
5. Food preservation and cold-storage technical literature.
6. Relevant academic papers and institutional/FAO resources on post-harvest handling.

Acknowledgement: Special thanks to the faculty, department and institution for their guidance and support in developing this project.

Smart Technology + Natural Cooling = Sustainable Vegetable Preservation